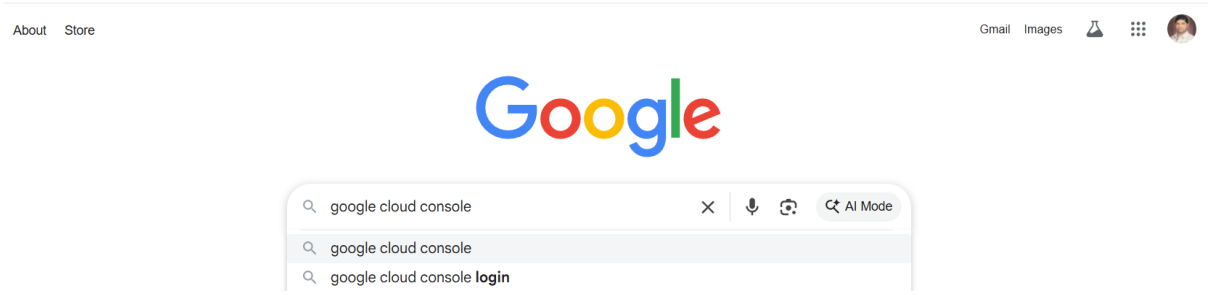
Creating OAuth for Client Side

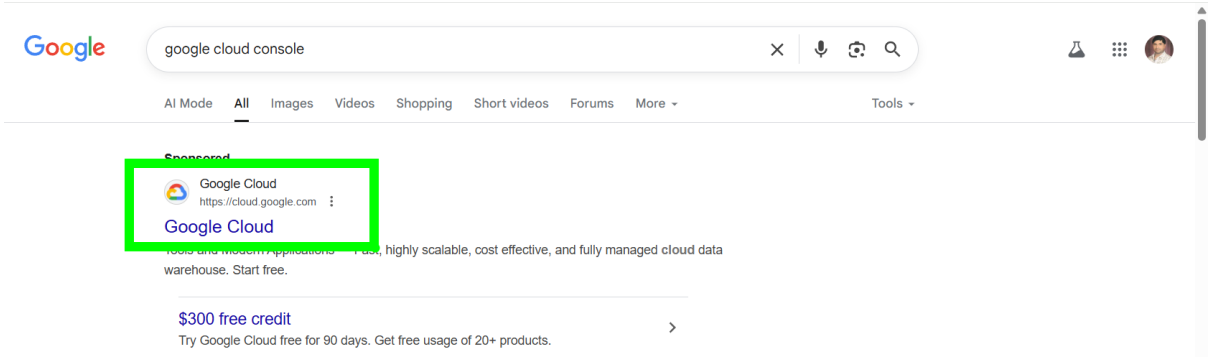
Note: The highlighted Green Color Box indicates you need to click on it

- **Step 1: Setup in Google Cloud Console**

1. Go to Google Cloud Console

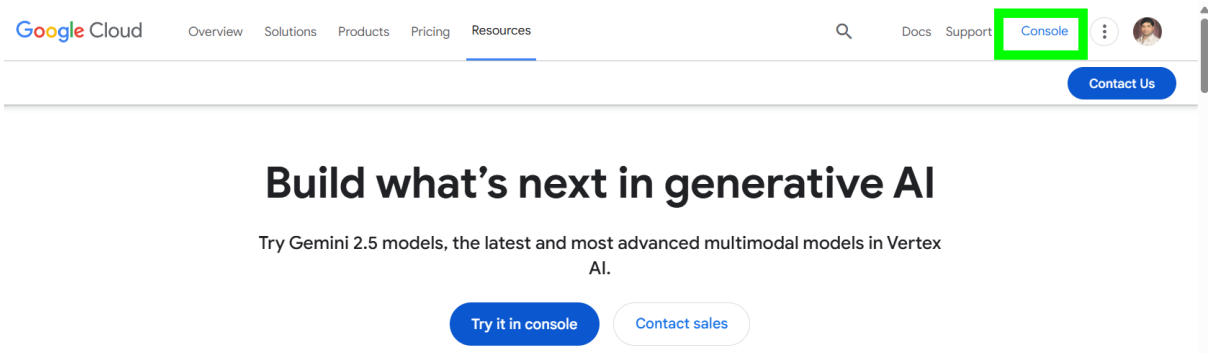


Click on Google Cloud



If you are already logged in using your google signin then the below page will be displayed.

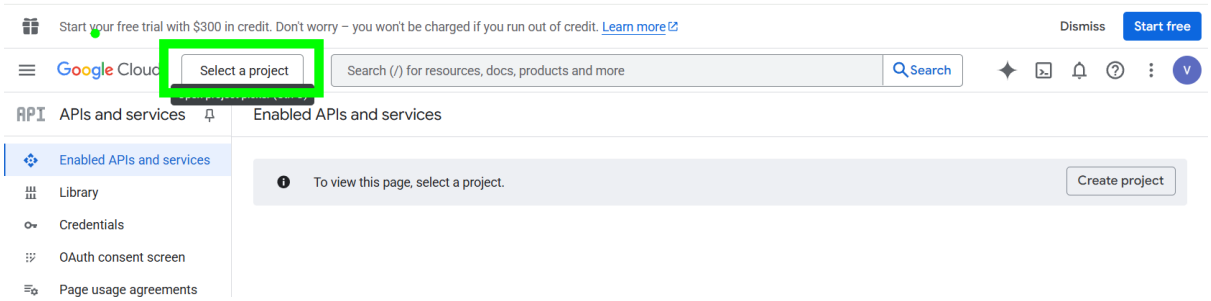
Click on Console



You will be redirected to the Google cloud page displayed below:

If you are new to Google Cloud then this page will be displayed.

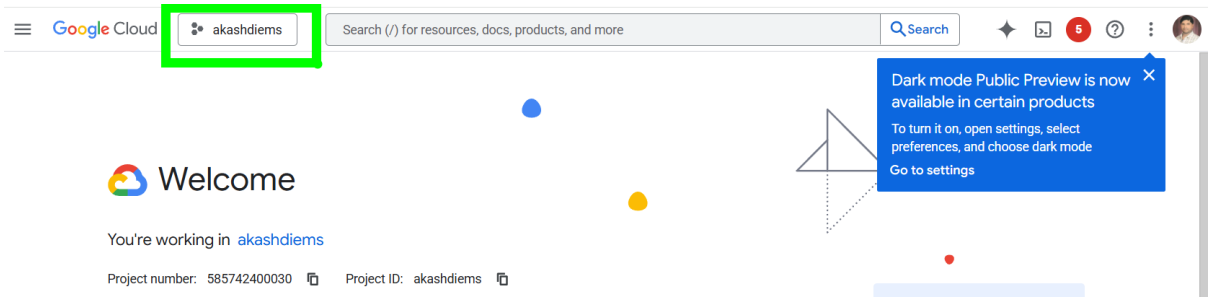
Click on the “Select a Project”



(OR)

If you have already accessed to Google Cloud then this page will be displayed where you will see some project name in this case “akashdiems” is already created project.

Click on the project Name

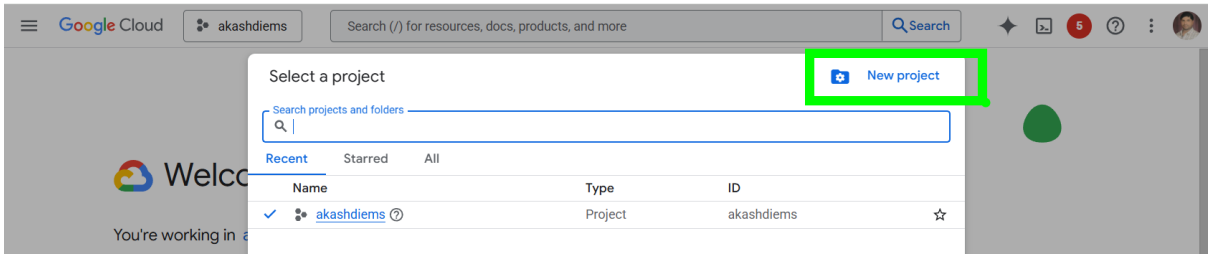


If you have clicked on:

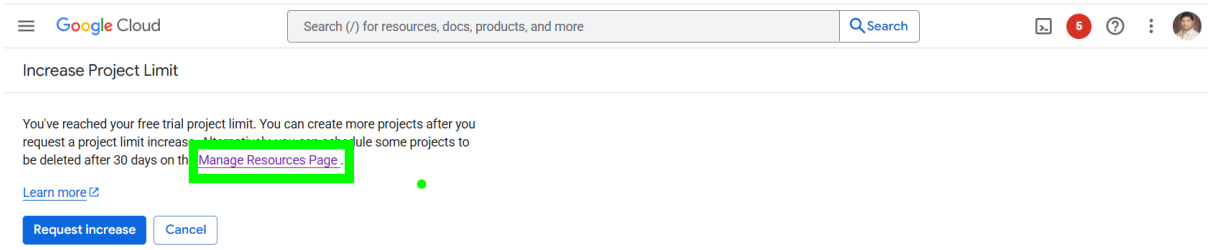
Click on the “Select a Project”

Or

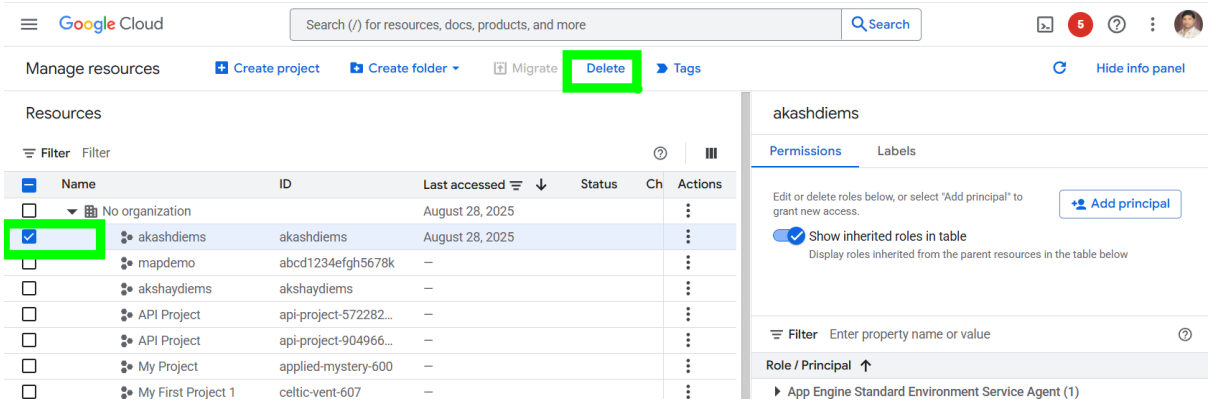
Click on Project Name “akashdiems”

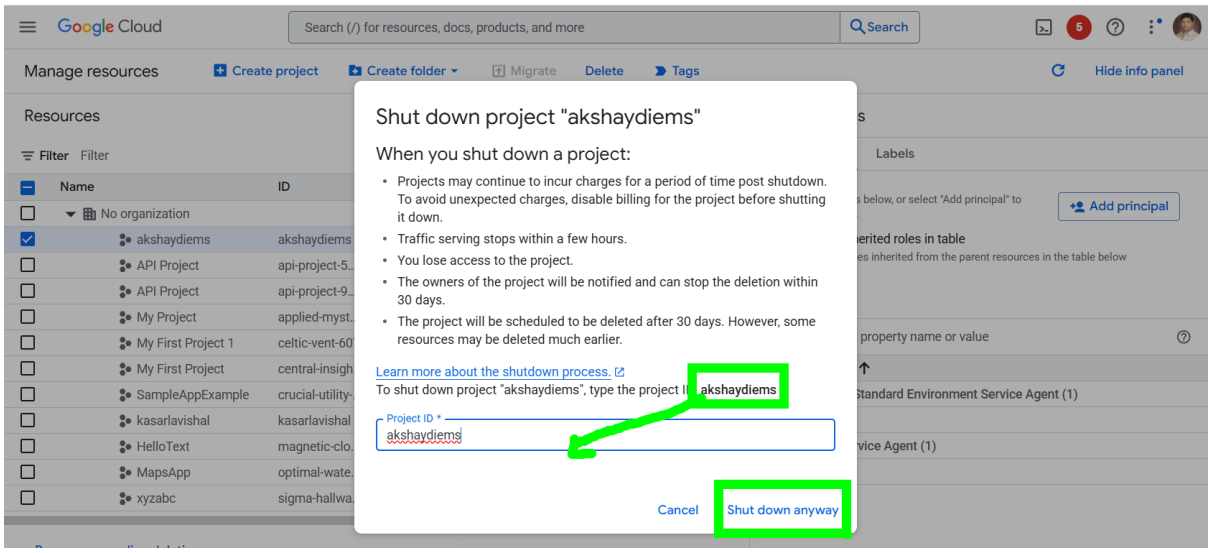


If you have exceeded the Project creation limit then this **“error”** on the webpage then click on Manage Resource Page



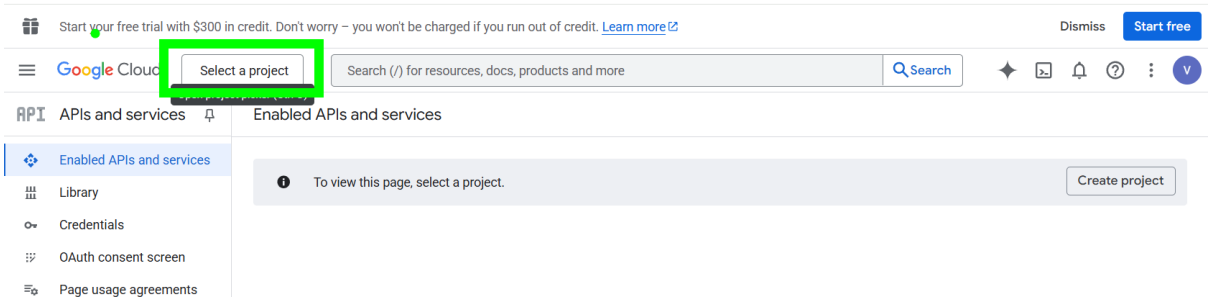
Select the projects to be deleted



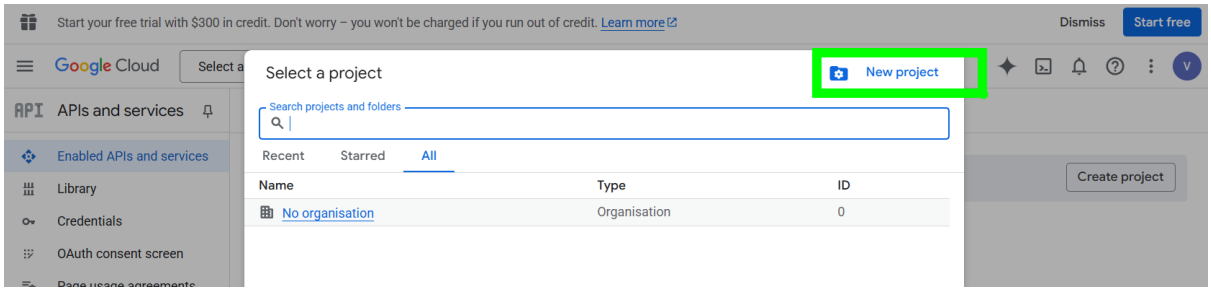


This will take 30 days for deleting an existing project.

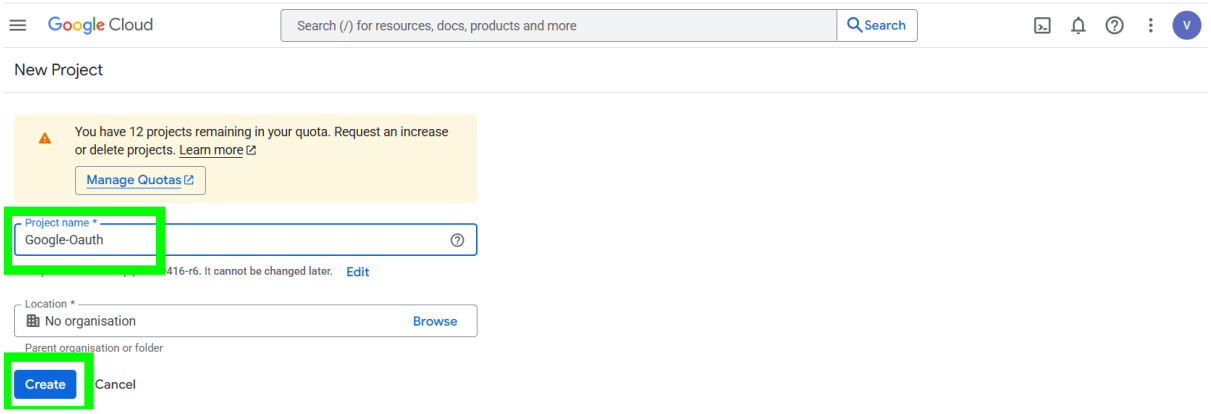
Instead you can use an existing project or else if you do not have any existing projects then create a new project



Click on the New Project

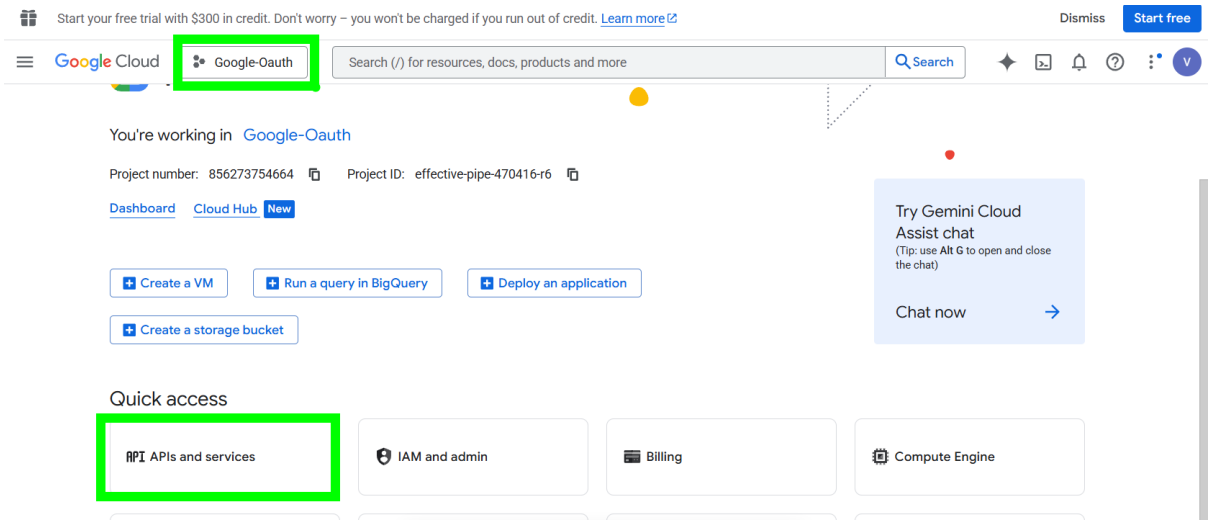


Name your Project – In this case “Google-Oauth” then click on Create.

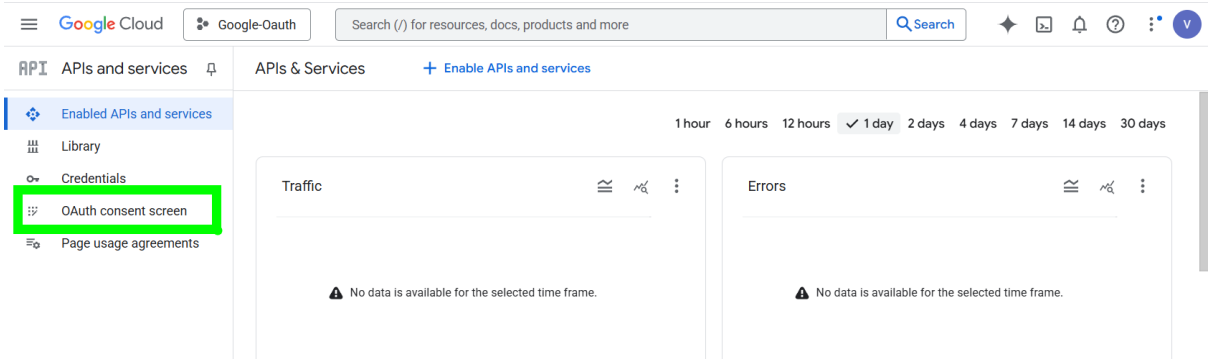


2. Create a new **OAuth 2.0 Client ID**.

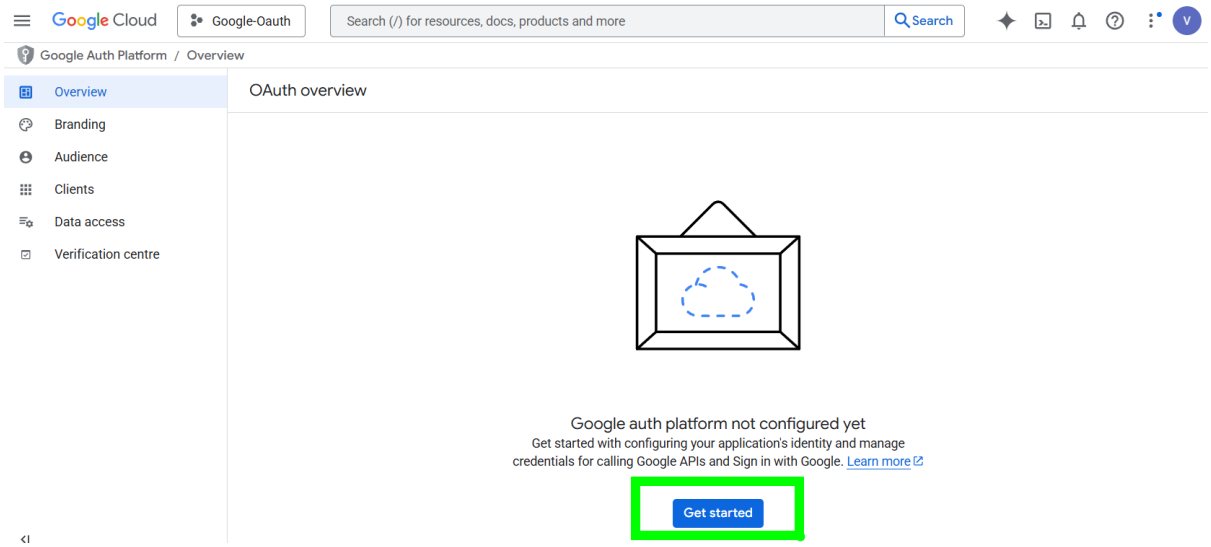
After Creating the project – Now we need to click on APIs and Services.



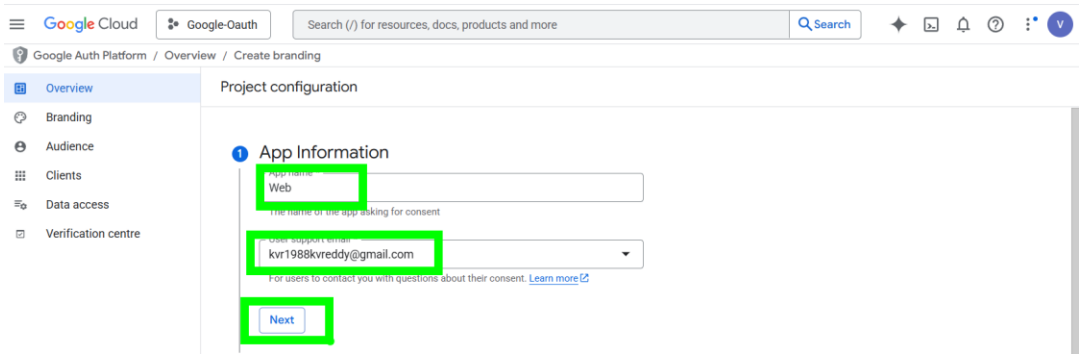
Click on OAuth Consent Screen – to create the Google Client ID

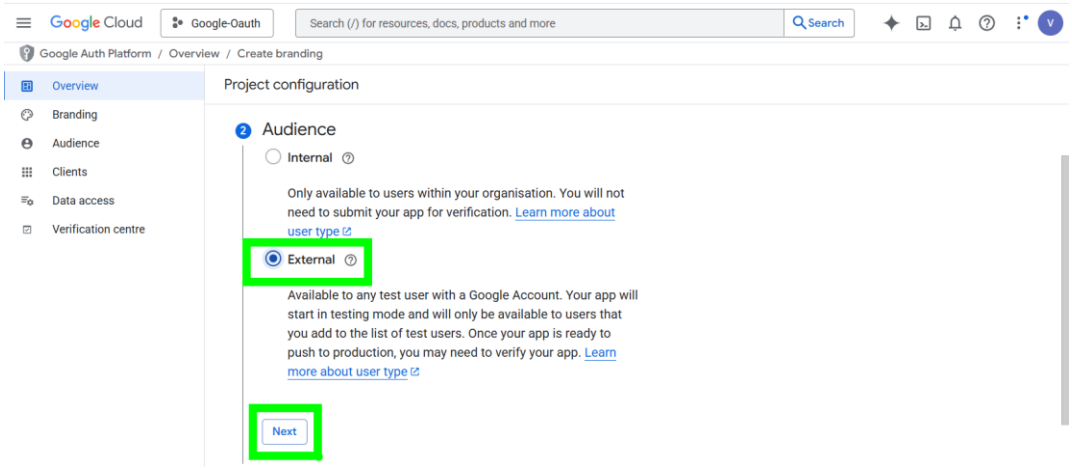


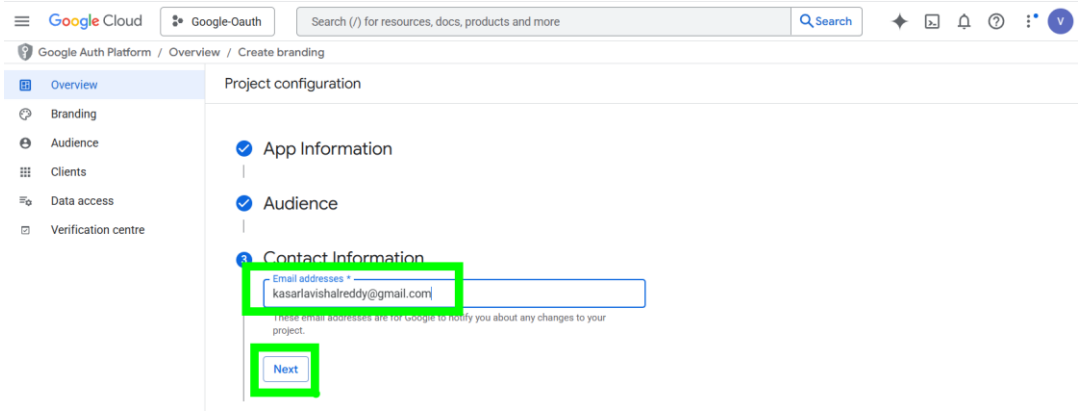
Click on Get Started

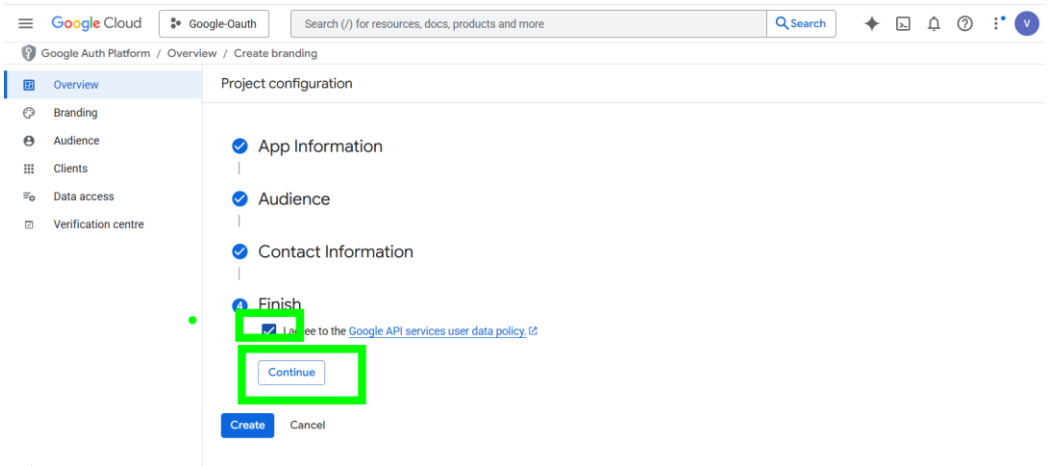


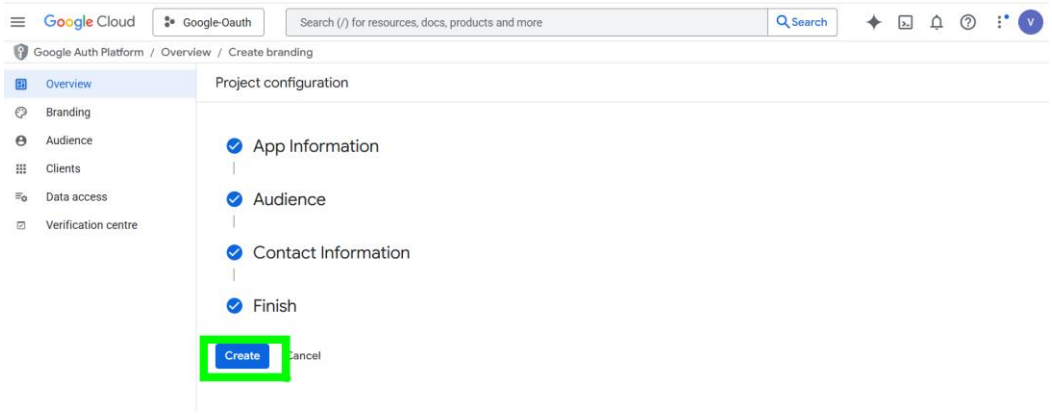
Provide App Information – Complete 5 Steps as mentioned below (By Scrolling Down)



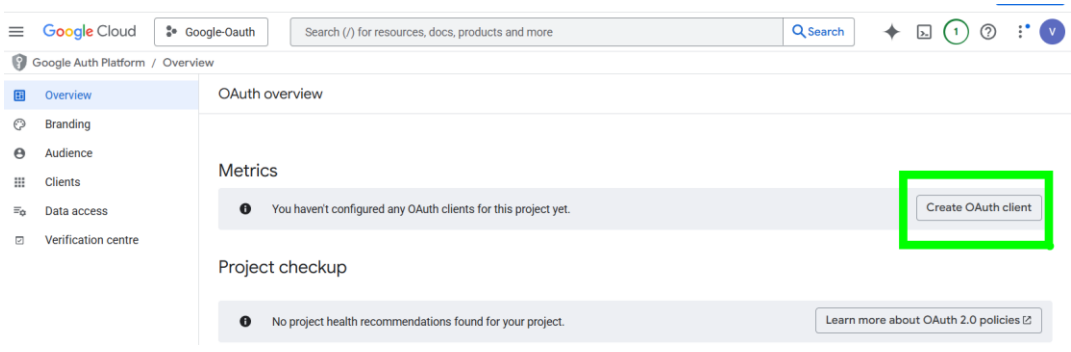


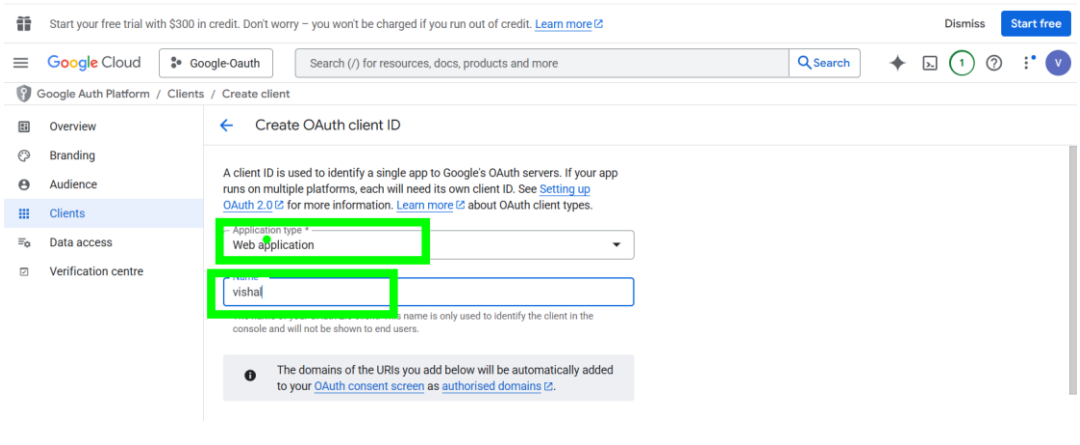


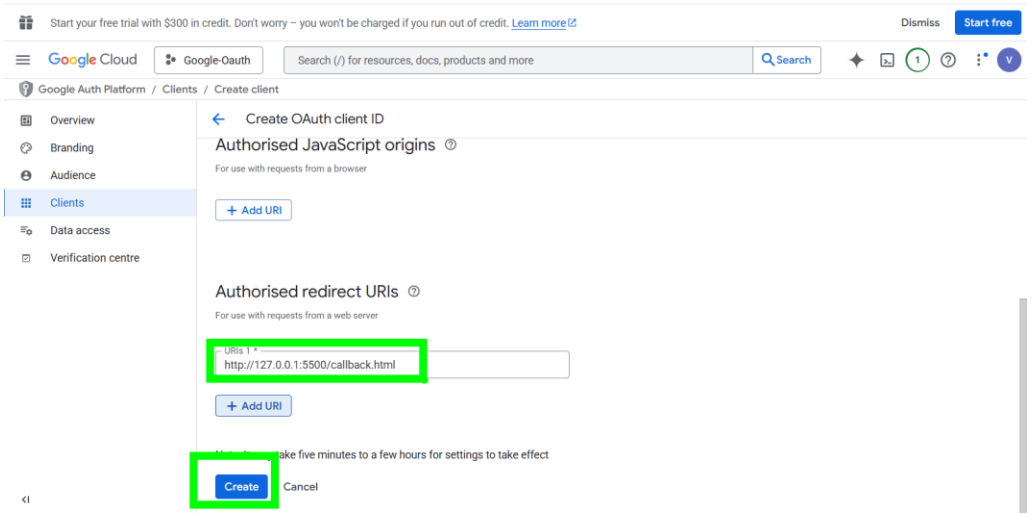




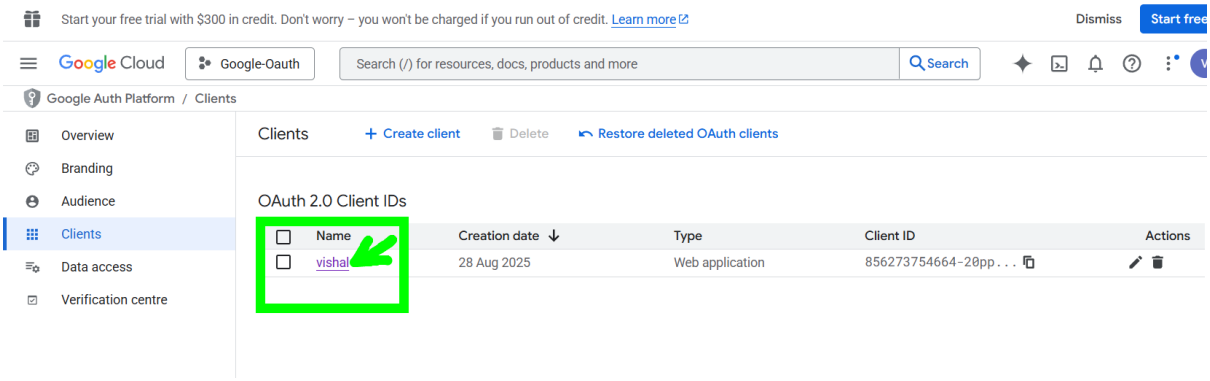
Now click on Create OAuth Client



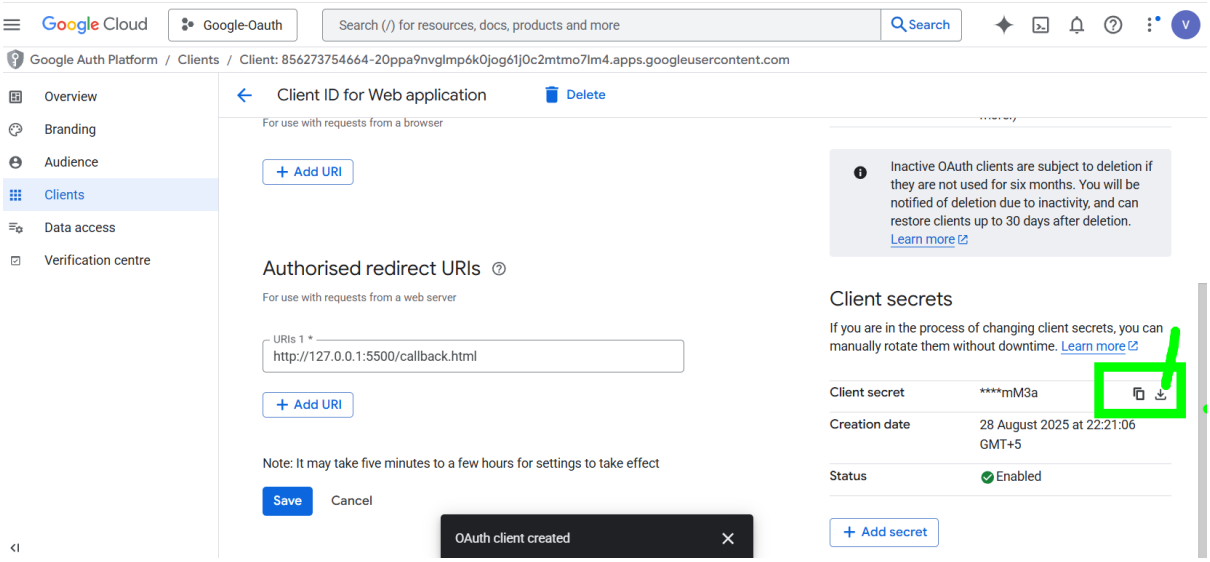




Once the Google Client ID has Created – then click the Name

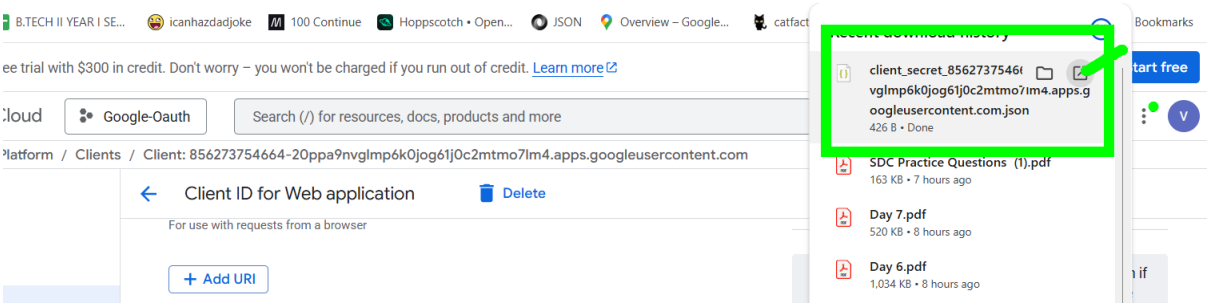


Scroll down under Client Secrets click on the Download symbol.

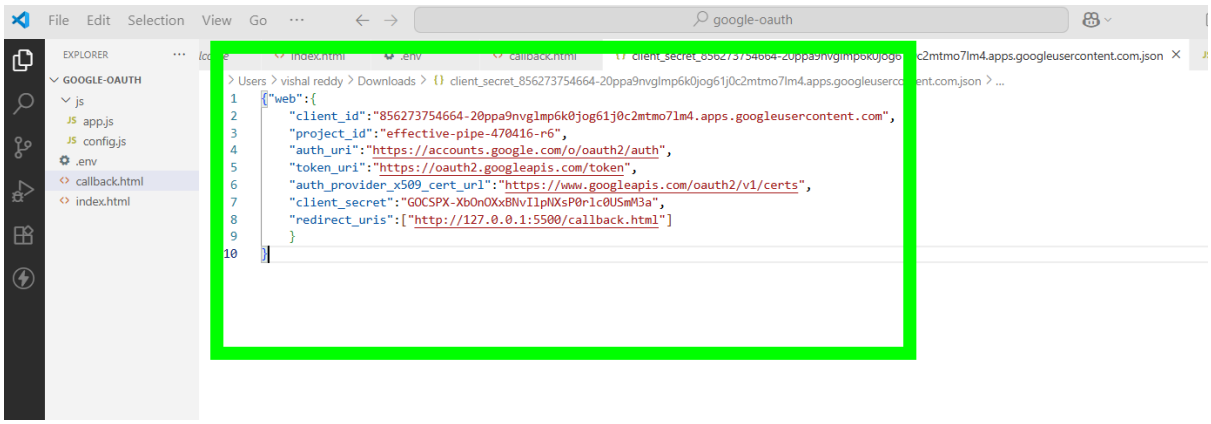


The JSON file gets Downloaded .. Open the JSON file to get

Client ID, Redirect Urls, and so on…



Now Copy your **Client ID and and** redirect URL



- **Step 2: Set Up Project Folder**
  1. Create a folder google-oauth-demo/
  2. Inside it, create files like this:

📁 Application Structure

google-oauth-demo/

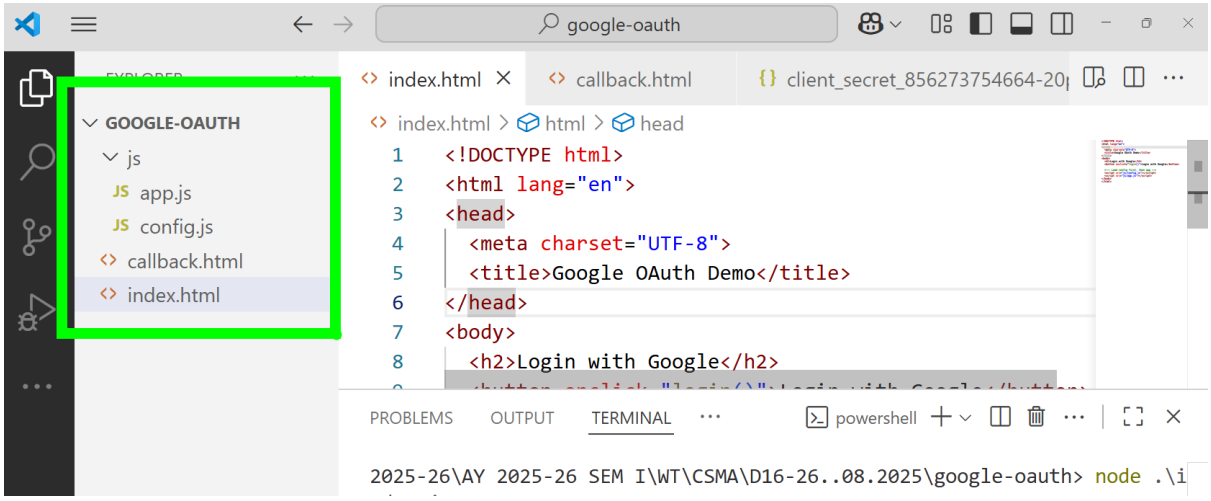
|— index.html

|— callback.html

|— js/

| ├— config.js (contains Client ID, Redirect URI)

| └— app.js (main JS logic)



◇ 1. js/config.js

```
// Configuration file for OAuth
const GOOGLE_CLIENT_ID = "PASTE_YOUR_CLIENT_ID.apps.googleusercontent.com";
const GOOGLE_REDIRECT_URI = "http://127.0.0.1:5500/callback.html";
const GOOGLE_SCOPE = "openid profile email";
```

◇ 2. js/app.js

```
// Handle login redirection
function login() {
  const authUrl = `https://accounts.google.com/o/oauth2/v2/auth?response_type=token&client_id=${GOOGLE_CLIENT_ID}&redirect_uri=${GOOGLE_REDIRECT_URI}&scope=${GOOGLE_SCOPE}`;
  window.location = authUrl;
}
```

◇ 3. index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>Google OAuth Demo</title>
</head>
<body>
  <h2>Login with Google</h2>
  <button onclick="login()">Login with Google</button>

  <!-- Load config first, then app -->
  <script src="js/config.js"></script>
  <script src="js/app.js"></script>
</body>
```

```
</html>
```

◇ 4. callback.html

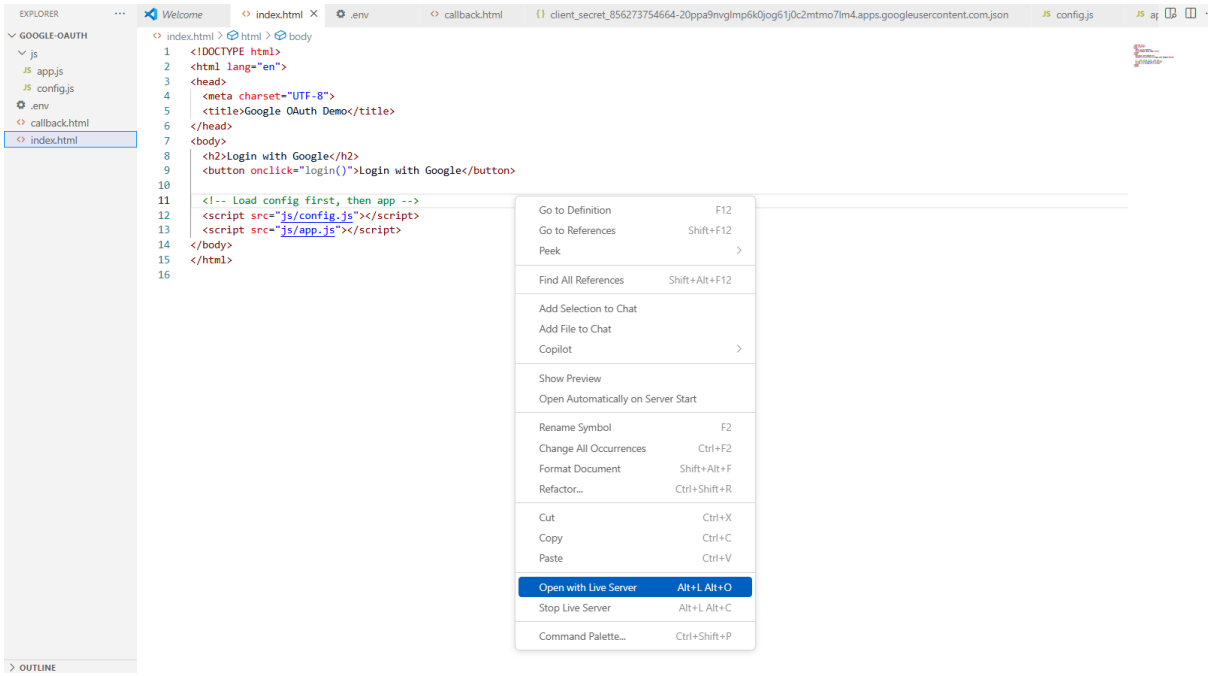
```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <title>OAuth Callback</title>
</head>
<body>
  <h2>OAuth Callback</h2>
  <div id="result"></div>

  <script>
    // Extract token from URL hash
    const hash = window.location.hash.substr(1);
    const params = new URLSearchParams(hash);
    const accessToken = params.get("access_token");

    if (accessToken) {
      fetch("https://www.googleapis.com/oauth2/v3/userinfo", {
        headers: { Authorization: "Bearer " + accessToken }
      })
      .then(res => res.json())
      .then(user => {
        document.getElementById("result").innerHTML = `
          <h3>Welcome, ${user.name}</h3>
          <p>Email: ${user.email}</p>
          <img src="${user.picture}" width="100"/>
        `;
```

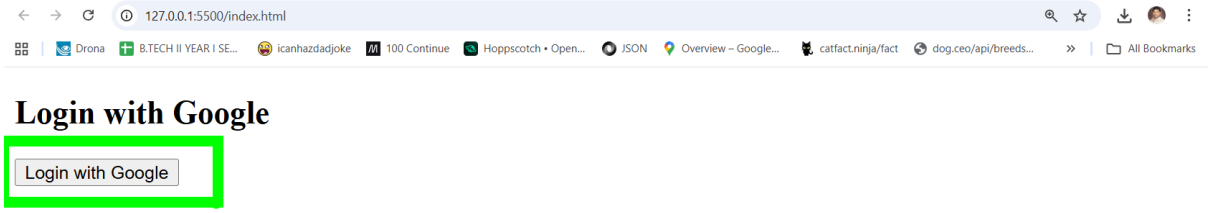
```
  });
 } else {
  document.getElementById("result").innerText = "No access token found!";
 }
</script>
</body>
</html>
```

- **Step 3: Open in VS Code**
  1. Open **VS Code** → File → Open Folder → google-oauth-demo.
  2. Install **Live Server extension** in VS Code (if not already).
- **Step 4: Run with Live Server**
  1. Right-click index.html → **"Open with Live Server"**.



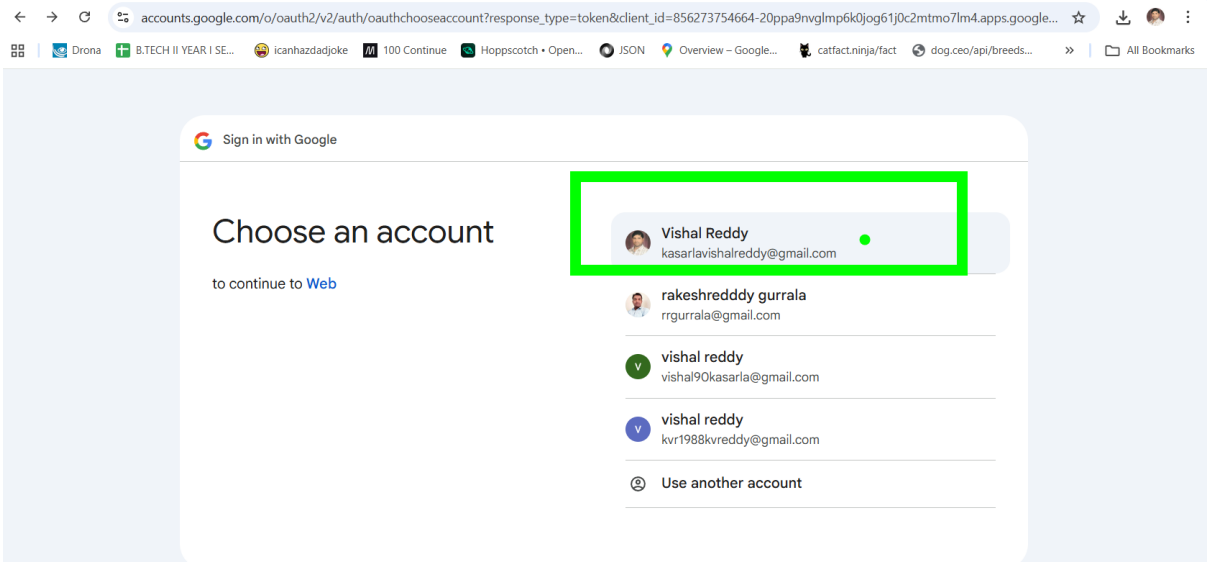
2. It will open in your browser at:

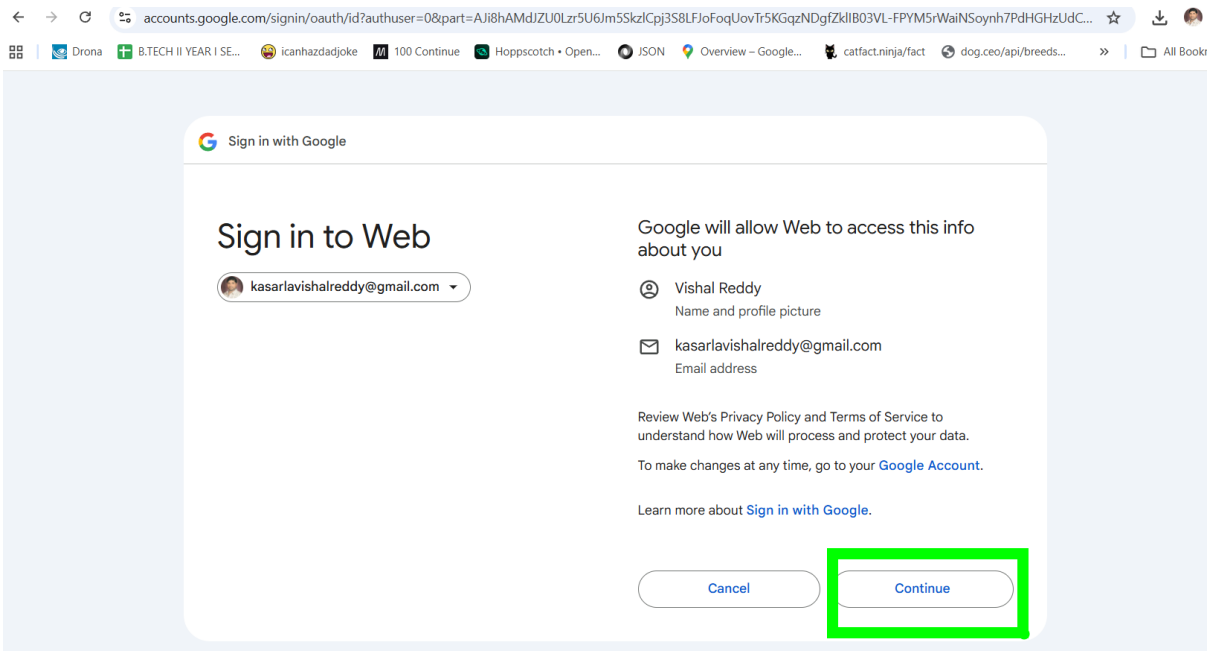
   http://127.0.0.1:5500/index.html



3. Click **Login with Google**.

4. You'll be redirected to Google's login screen.





5. After login, Google will redirect to:

http://127.0.0.1:5500/callback.html#access_token=...

6. The page will fetch your Google profile info and display:

- ✅ Name
- ✅ Email
- ✅ Profile Picture



- **Workflow**

1. index.html → loads config.js + app.js.
2. Click login → redirect to Google OAuth.
3. Google redirects back to callback.html#access_token=....
4. callback.html fetches Google **userinfo API** with access_token.
5. Displays user details.